

Store Table

Q1 Create the store table with two column families: details and inventory.

```
2 row(s)
Took 1.8098 seconds
=> ["emp", "student"]
hbase(main):002:0> create 'store', 'details', 'inventory'
Created table store
Took 1.3622 seconds
=> Hbase::Table - store
hbase(main):003:0> list
TABLE
emp
store
student
3 row(s)
Took 0.0236 seconds
=> ["emp", "store", "student"]
hbase(main):004:0> █
```

Q3 Scan the entire store table and confirm all 5 rows are present.

```
Took 0.0249 seconds
hbase(main):038:0> scan 'store'
ROW COLUMN+CELL
1 column=details:brand, timestamp=2026-05-24T12:52:20.431, value=dell
1 column=details:category, timestamp=2026-05-24T12:52:20.540, value=electronics
1 column=details:name, timestamp=2026-05-24T12:52:20.259, value=laptop
1 column=inventory:price, timestamp=2026-05-24T12:52:20.668, value=1200
1 column=inventory:quantity, timestamp=2026-05-24T12:52:20.763, value=50
1 column=inventory:warehouse, timestamp=2026-05-24T12:52:20.840, value=A
2 column=details:brand, timestamp=2026-05-24T12:52:21.051, value=samsung
2 column=details:category, timestamp=2026-05-24T12:52:21.119, value=electronics
2 column=details:name, timestamp=2026-05-24T12:52:20.961, value=phone
2 column=inventory:price, timestamp=2026-05-24T12:52:21.182, value=800
2 column=inventory:quantity, timestamp=2026-05-24T12:52:21.241, value=120
2 column=inventory:warehouse, timestamp=2026-05-24T12:52:21.360, value=A
3 column=details:brand, timestamp=2026-05-24T12:52:21.589, value=ikea
3 column=details:category, timestamp=2026-05-24T12:52:21.649
```

Q4 Retrieve all columns for the product with row key 3 (desk).

```
Took 0.3151 seconds
hbase(main):039:0> get 'store', '3'
COLUMN CELL
details:brand timestamp=2026-05-24T12:52:21.589, value=ikea
details:category timestamp=2026-05-24T12:52:21.649, value=furniture
details:name timestamp=2026-05-24T12:52:21.457, value=desk
inventory:price timestamp=2026-05-24T12:52:21.698, value=300
inventory:quantity timestamp=2026-05-24T12:52:21.796, value=30
inventory:warehouse timestamp=2026-05-24T12:52:22.127, value=B
1 row(s)
Took 0.0617 seconds
hbase(main):040:0> █
```

Q5 Retrieve only the inventory:price column for the product with row key 5 (tablet).

```
Took 0.0369 seconds
hbase(main):040:0> get 'store', '5', {COLUMN => 'inventory:price'}
COLUMN CELL
inventory:price timestamp=2026-05-24T12:52:22.655, value=600
1 row(s)
Took 0.0369 seconds
hbase(main):041:0> █
```

Q6 Use a ValueFilter scan to find all products with warehouse value A.

```
took 0.0369 seconds
hbase(main):041:0> scan 'store', {
hbase(main):042:1* COLUMNS => ['inventory:warehouse'],
hbase(main):043:1* FILTER => "ValueFilter(=,'binary:A')"
hbase(main):044:1> }
ROW                                COLUMN+CELL
1                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:20.840, value=A
2                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:21.360, value=A
5                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:35.570, value=A
3 row(s)
Took 0.2491 seconds
hbase(main):045:0> █
```

PART B — INTERMEDIATE

Q7 Enable versioning on the inventory column family to store up to 4 versions.

```
hbase(main):045:0> alter 'store', NAME => 'inventory', VERSIONS => 4
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 2.5744 seconds
hbase(main):046:0> █
```

Q8 Update the inventory:quantity for the laptop (row 1) three times: first to 45, then to 40, then to 35. Then retrieve all stored versions of that cell.

```
hbase(main):049:0>
hbase(main):050:0> get 'store', '1', {COLUMN => 'inventory:quantity', VERSIONS => 4}
COLUMN                                CELL
inventory:quantity                    timestamp=2026-05-24T13:02:51.583, value=35
inventory:quantity                    timestamp=2026-05-24T13:02:51.533, value=40
inventory:quantity                    timestamp=2026-05-24T13:02:51.474, value=45
inventory:quantity                    timestamp=2026-05-24T12:52:20.763, value=50
1 row(s)
Took 0.0560 seconds
hbase(main):051:0> █
```

Q9 Use SingleColumnValueFilter to find all products in the furniture category.

```
hbase(main):051:0> scan 'store', {
hbase(main):052:1* FILTER => "SingleColumnValueFilter('details','category',=,'binary:furniture')"
hbase(main):053:1> }
ROW                                COLUMN+CELL
3                                  column=details:brand, timestamp=2026-05-24T12:52:21.589, value=ikea
3                                  column=details:category, timestamp=2026-05-24T12:52:21.649, value=furniture
3                                  column=details:name, timestamp=2026-05-24T12:52:21.457, value=desk
3                                  column=inventory:price, timestamp=2026-05-24T12:52:21.698, value=300
3                                  column=inventory:quantity, timestamp=2026-05-24T12:52:21.796, value=30
3                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:22.127, value=B
4                                  column=details:brand, timestamp=2026-05-24T12:52:22.257, value=ikea
4                                  column=details:category, timestamp=2026-05-24T12:52:22.318, value=furniture
4                                  column=details:name, timestamp=2026-05-24T12:52:22.204, value=chair
4                                  column=inventory:price, timestamp=2026-05-24T12:52:22.380, value=150
4                                  column=inventory:quantity, timestamp=2026-05-24T12:52:22.422, value=60
4                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:22.467, value=B
row(s)
Took 0.0853 seconds
```

Q10 Use SingleColumnValueFilter to find all products stored in warehouse B.

```
hbase(main):055:1* FILTER => "SingleColumnValueFilter('inventory','warehouse',=,'binary:B')"
hbase(main):056:1> }
ROW                                COLUMN+CELL
3                                  column=details:brand, timestamp=2026-05-24T12:52:21.589, value=ikea
3                                  column=details:category, timestamp=2026-05-24T12:52:21.649, value=furniture
3                                  column=details:name, timestamp=2026-05-24T12:52:21.457, value=desk
3                                  column=inventory:price, timestamp=2026-05-24T12:52:21.698, value=300
3                                  column=inventory:quantity, timestamp=2026-05-24T12:52:21.796, value=30
3                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:22.127, value=B
4                                  column=details:brand, timestamp=2026-05-24T12:52:22.257, value=ikea
4                                  column=details:category, timestamp=2026-05-24T12:52:22.318, value=furniture
4                                  column=details:name, timestamp=2026-05-24T12:52:22.204, value=chair
4                                  column=inventory:price, timestamp=2026-05-24T12:52:22.380, value=150
4                                  column=inventory:quantity, timestamp=2026-05-24T12:52:22.422, value=60
4                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:22.467, value=B
2 row(s)
Took 0.0781 seconds
hbase(main):057:0> █
```

Q11 Add a new column family called supplier to the store table. Then insert supplier:country as egypt for row 1 and supplier:country as korea for row 2. Verify the schema updated correctly.

```
hbase(main):058:0>
hbase(main):059:0> put 'store', '1', 'supplier:country', 'egypt'
Took 0.0262 seconds
hbase(main):060:0> put 'store', '2', 'supplier:country', 'korea'
Took 0.0062 seconds
hbase(main):061:0>
hbase(main):062:0> describe 'store'
Table store is ENABLED
store
COLUMN FAMILIES DESCRIPTION
{NAME => 'details', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}
{NAME => 'inventory', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '4', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}
{NAME => 'supplier', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}
3 row(s)
QUOTAS
0 row(s)
Took 0.1430 seconds
hbase(main):063:0>
```

Q12 CHALLENGE 🏆 Write a single scan that returns only products that are in the electronics category AND stored in warehouse A.

```
0 row(s)
Took 0.1430 seconds
hbase(main):063:0> scan 'store', {
hbase(main):064:1* FILTER => "SingleColumnValueFilter('details','category',=,'binary:electronics') AND SingleColumnValueFilter('inventory','warehouse',=,'binary:A')"}
hbase(main):065:1> }
ROW                                COLUMN+CELL
1                                  column=details:brand, timestamp=2026-05-24T12:52:20.431, value=dell
1                                  column=details:category, timestamp=2026-05-24T12:52:20.540, value=electronics
1                                  column=details:name, timestamp=2026-05-24T12:52:20.259, value=laptop
1                                  column=inventory:price, timestamp=2026-05-24T12:52:20.668, value=1200
1                                  column=inventory:quantity, timestamp=2026-05-24T13:02:51.583, value=35
1                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:20.840, value=A
1                                  column=supplier:country, timestamp=2026-05-24T13:09:04.606, value=egypt
2                                  column=details:brand, timestamp=2026-05-24T12:52:21.051, value=samsung
2                                  column=details:category, timestamp=2026-05-24T12:52:21.119, value=electronics
2                                  column=details:name, timestamp=2026-05-24T12:52:20.961, value=phone
2                                  column=inventory:price, timestamp=2026-05-24T12:52:21.182, value=800
2                                  column=inventory:quantity, timestamp=2026-05-24T12:52:21.241, value=120
2                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:21.360, value=A
2                                  column=supplier:country, timestamp=2026-05-24T13:09:04.663, value=korea
5                                  column=details:brand, timestamp=2026-05-24T12:52:22.575, value=apple
5                                  column=details:category, timestamp=2026-05-24T12:52:22.623, value=electronics
5                                  column=details:name, timestamp=2026-05-24T12:52:22.528, value=tablet
5                                  column=inventory:price, timestamp=2026-05-24T12:52:22.655, value=600
5                                  column=inventory:quantity, timestamp=2026-05-24T12:52:22.693, value=80
5                                  column=inventory:warehouse, timestamp=2026-05-24T12:52:35.570, value=A
3 row(s)
Took 0.1200 seconds
hbase(main):066:0>
```